SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5109
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:5 **Total Marks:100**

Annexure A

Code of Conduct:

I K.Mounith(192425090) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Mounith

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n ϕ(n)\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

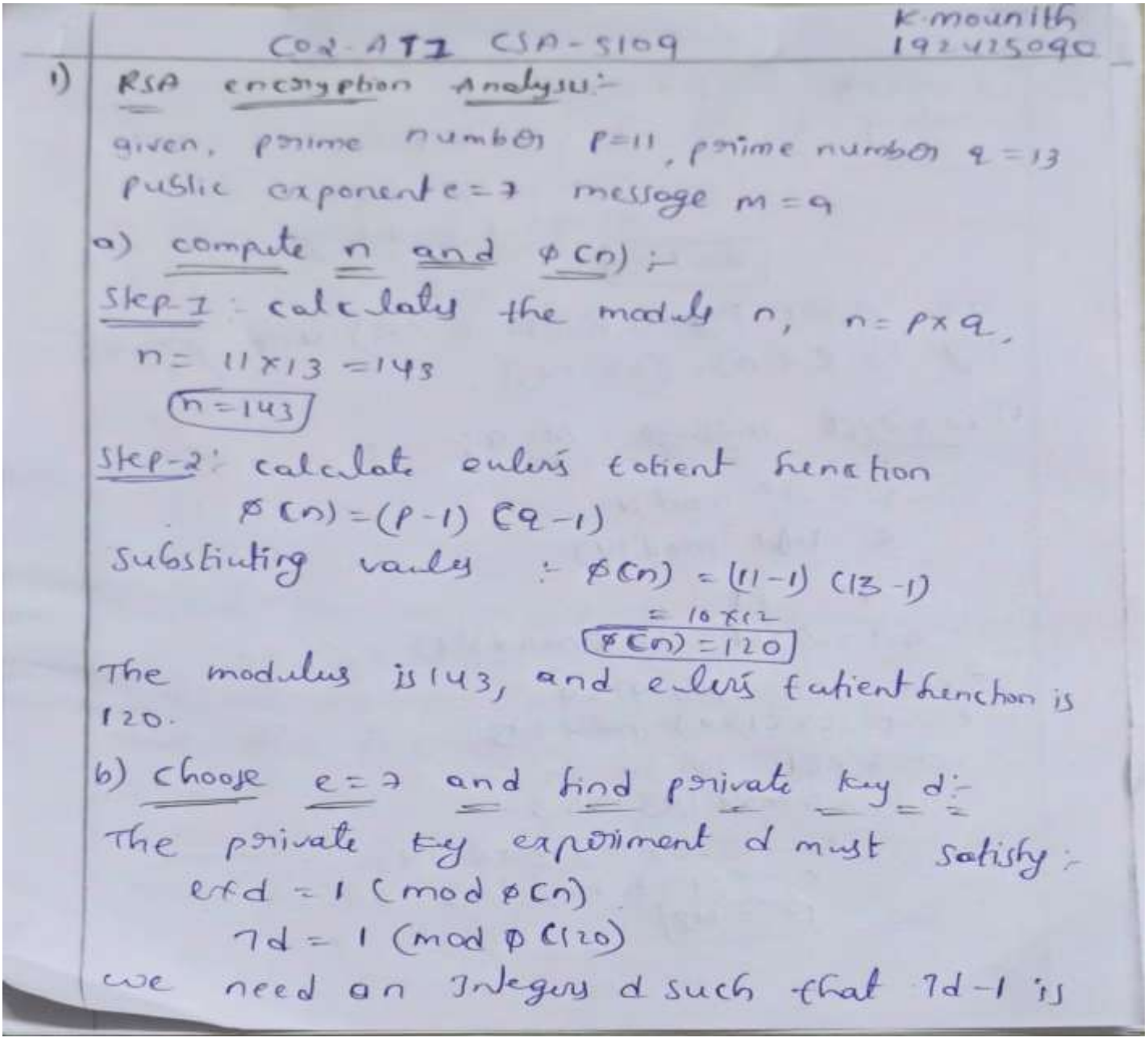
4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

ANSWERS:



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1) RSA encryption Analysis:-

given, prime number p=11, prime number q=13
public exponent e=7 message m=9

a) compute n and $\phi(n)$:-

Step-1: calculate the modulus n, $n = p \times q$,

$n = 11 \times 13 = 143$

$\boxed{n = 143}$

Step-2: calculate eulers totient function

$\phi(n) = (p-1)(q-1)$

Substituting values: $\phi(n) = (11-1)(13-1)$

$= 10 \times 12$

$\boxed{\phi(n) = 120}$

The modulus is 143, and eulers totient function is 120.

b) Choose e=7 and find private key d:-

The private key exponent d must satisfy:-

$ed = 1 \pmod{\phi(n)}$

$7d = 1 \pmod{\phi(120)}$

we need an Integers d such that 7d-1 is

divisable by 120

taking $d = 103$ $\quad 7 \times 103 = 721$

$721 = 6 \times 120 + 1$

$7 \times 103 = 1 \pmod{120}$

$\boxed{d = 103}$

The public key is $(e, n) = (7, 143)$ and private key is $(d, n) = (103, 143)$.

c) encrypt message $M = 9$:-

$C = M^e \bmod N$

$C = 9^7 \bmod 143$

$9^2 = 81$

$9^4 = 6561 = 126 \pmod{143}$

Since $9^7 = 9^4 \times 9^2 \times 9$

$C = (126 \times 81 \times 9) \bmod 143$

$126 \times 81 = 10206$

$10206 \bmod 143 = 53$

$C = (53 \times 9) \bmod 143$

$C = 477 \bmod 143$

$\boxed{C = 48}$

d) Decrypt the ciphertext and verify the result.

RSA decryption uses: $M = c^d \bmod n$

$M = 48^{143} \bmod 143$

power	value modulo 143
48^1	48
48^2	16
48^4	113
48^4	42
48^{16}	48
48^{32}	16
48^{64}	113

Express the exponent as: $64+32+4+2+1 = 103$.

$M = 113 \times 16 \times 113 \times 16 \times 48 \pmod{143}$ $m = 9$

original message = 9 = decrypted message

Thus, RSA encryption and decryption are success.

e) Analyze how changing e affects security

The public exponent e must satisfy:

$1 < e < \phi(n)$ &

$\gcd(e, \phi(n)) = 1$

In practical RSA systems $e = 65537$, is commonly used with adequately sized key and secure padding.

2) Block cipher mode evaluation

Given

plaintext blocks : $P = [1010, 1010, 1111, 1010]$

encryption key : $k = 1100$

Initialization vector $IV = 0000$

for this calculation, assume a simplified 4-bit block cipher : $E_k(x) = x \oplus k$

a) encrypt using ECB and CBC modes:

ECB (electronic code book) mode:

using assumed cipher: $c_i = p_i \oplus k$

formula: $\boxed{c_i = E_k(p_i)}$

Block 0: $c_i = 1010 \oplus 1100$ Block 1: $C_1 = 1010 \oplus 1100$

$\boxed{C_1 = 0110}$

$C_1 = 0110$

Block 3:- $C_3 = 1111 \oplus 1100$
$= 0011$

Block 4:- $C_4 = 1010 \oplus 1100$

$\boxed{C_4 = 0110}$

Therefore the ECB ciphertext is

$C_{ECB} = [0110, 0110, 0011, 0110]$

∴ CBC ciphertext is

$C_{CBC} = [0110, 0000, 0011, 0101]$

b) Compare ciphertext outputs:-

Feature	ECB mode:-	CBC mode:-
1st block	0110	0110
2nd block	0110	0110
3rd block	0011	0011
4th block	0110	0101
Identical plaintext	produce Identical	
chaning errors	Cipher text not	
propogation	used generally	
	limited to affected data.	

ECB ciphertext:- 0110 0110 0011 0110

CBC ciphertext:- 0110 0000 0011 0101.

c) Analyze which mode hides patterns better.

CBC generally hides repeated plaintext patterns better than ECB because each block depends on the previous ciphertext block, with a secure block cipher and a properly generated unpredictable IV, repeated plain text blocks at different positions can product different ciphertext blocks.

3) DES vs 3-DES vs Blowfish performance study:

Given:

Algorithm	Key size	encryption Time/block
DES	56 bits	2ms
3-DES	168 bits	6ms
Blowfish	128 bits	1ms

no. of data blocks $N = 100$.

i) calculate total encryption for each algorithm

$T = N \times t$

T → total encryption, N → no of blocks, t → encryption / block.

A) DES $T_{DES} = 1000 \times 2$

$T_{DES} = 2000ms = 2 sec$

B) 3-DES $T_{3-DES} = 1000 \times 6$

$T_{3-DES} = 6000ms = 6 seconds.$

C) Blowfish $T_{Blowfish} = 1000 \times 1$

$T_{Blowfish} = 1000ms = 1 sec$

Comparison:

Algorithm	keysize	Time per Block	Total Time for 1000 Blocks
DES	56 bits	2ms	2sec
3-DES	168 bits	6ms	6 secs
Blowfish	128 bits	1ms	1 sec

ii) Analyse trade-off - b/w security and performance:

* DES uses 56-bit effective key. Its small key size makes it vulnerable to brute-force attacks using modern computing capabilities.
* 3-DES applies the DES cipher three times.
* Blowfish supports variable key length and is efficient in given cases.

iii) Recommend the best algorithm for secure real-time communication:-

Blowfish is the fastest according to given data for new real communicating systems. It is more is generally more appropriate mostly DES and DES ahead be checked for new deployments.

4) Diffie-Hellman key exchange Analysis:-

given:- prime number $p = 23$

generate $g = 5$

user A's private key $a = 6$

user B's private key $b = 15$

a) compute the public key to user A:-

user A calculate public key using:-

$A = g^a \bmod p$.

$A = 5^6 \bmod 23$

calculate power $= 5^2 = 25 = 2 \pmod{23}$

$5^4 = 2^2 = 4 \pmod{23}$

$\therefore 5^6 = 5^4 \times 5^2 \pmod{23}$

$A = 4 \times 2 \pmod{23}$

$\boxed{A = 8}$

user A's public key is 8.

b) compute public key for user B:

user B calculate $B = g^b \bmod p = 5^{15} \bmod 23$

using repeated squaring $5^2 = 2 \pmod{23}$

$5^4 = 4 \pmod{23}$

$5^8 = 16 \pmod{23}$

express exponent $15 = 8+4+2+1$

$\therefore B = 5^8 \times 5^4 \times 5^2 \times 5 \pmod{23}$

$B = 16 \times 4 \times 2 \times 5 \pmod{23}$

Reducing mod 23: $16 \times 4 = 64 = 18$

$18 \times 2 = 36 = 13$

$13 \times 5 = 65 = 19$

user B's public key is 19.

c) compute shared secret key:

$K = B^a \bmod p$

$K = 19^6 \bmod 23$

$19^2 = 361 = 16 \pmod{23}$

$19^4 = 16^2 = 256 = 3$

$19^6 = 19^4 \times 19^2 \pmod{23}$

$K = 3 \times 16 \pmod{23}$

$\boxed{K = 2}$

user B independently calculate $K = A^b \bmod p$

$K = 8^{15} \bmod 23$

using repeated Squaring $8^2 = 18 \pmod{23}$

$8^4 = 18^2 = 2 \pmod{23}$

$8^8 = 2^2 = 4 \pmod{23}$

Since $15 = 8+4+2+1$

$K = 8^1 \times 8^4 \times 8^2 \times 8 \pmod{23}$

$K = 8^4 \times 2 \times 18 \times 8 \pmod{23}$

$K = 2 \pmod{23}$

$\boxed{K=2}$ Both users obtain same shared secret.

d) Demonstrate how a man-in the middle attack could alter exchange:

A man-in the middle (MITM) attack occurs when an attacker secretly intercepts and modifies public keys exchange b/w 2 users same attacker chooses private key $\boxed{x=3}$

$t = 5^3 \bmod 23$

$t = 125 \bmod 23$

$\boxed{t=10}$

Eve now shares secrete 6 with user A and secret 9 with user B $\boxed{K_{BE}=9}$

e) Suggest a secure method to prevent attack
* Digital certification
* Digital signature
* public-key verification
* secure protocols.

5) ECC vs RSA comparision:

a) Computational cost :- RSA uses larger keys & more computational resources, while ECC uses smaller keys.

b) ECC requires less memory and generally consumes less power.

c) Best choice for IoT :- ECC is better for IoT devices because it provides security with smaller keys and lower resources usage.

Conclusion: ECC is more suitable for IoT devices than RSA

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/form ula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculation s
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation